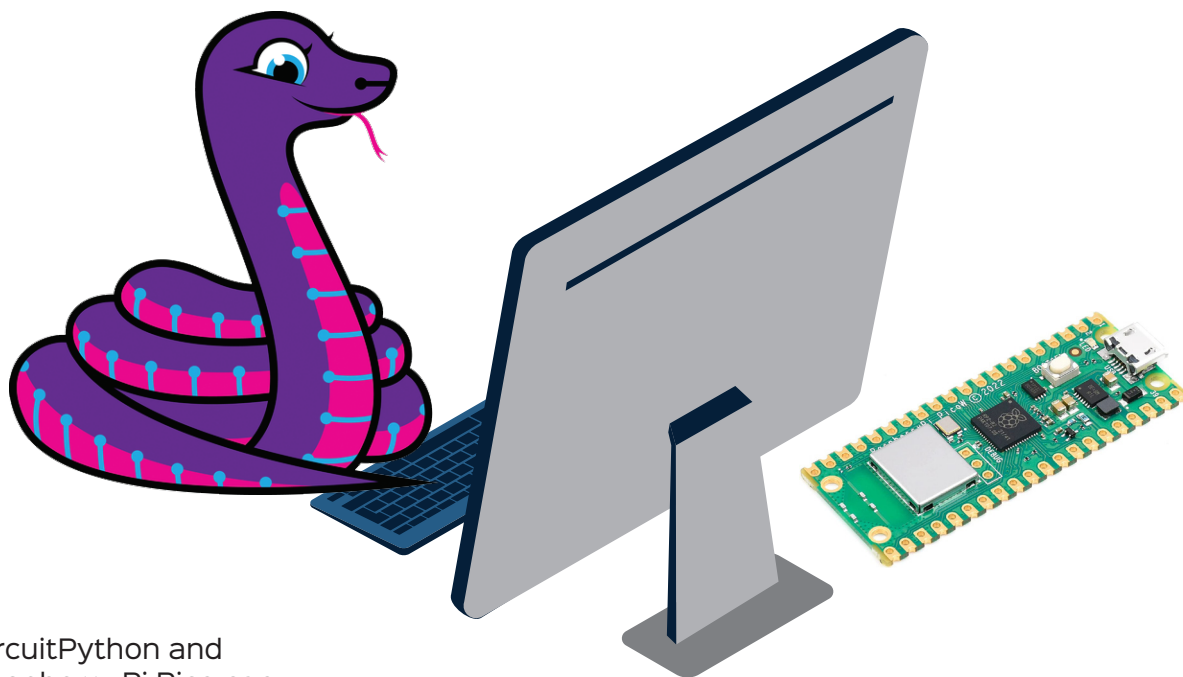


Leveraging the Power of CircuitPython for IoT Devices



CircuitPython and Raspberry Pi Pico can together transform everyday devices into smart wonders. This short tutorial will help you write a simple program for using CircuitPython and upload it to Raspberry Pi Pico.

The Internet of Things (IoT) refers to a network of devices capable of communicating with each other and the internet using unique identifiers. These devices have processing capabilities, high computing power, sensing units, and powerful embedded systems, allowing for applications as varied as ‘smart devices’, ‘smart cities’, and ‘smart schools’. IoT connects everyday objects like lights, monitors, cars, and doors to the internet,

enabling the exchange of data and information among people, processes, and things. This data and information find applications in fields like healthcare, education, and agriculture. For instance, IoT can help monitor farming environments, prevent theft by tracking home activities, control machines in factories, and report power outages.

Some of the most popular IoT devices are Alexa and smartwatches, which are used for home automation and healthcare. These devices rely on hardware components called microcontroller units (MCUs) and microprocessor units (MPUs) to perform complex tasks. For example, Alexa uses the NXP i.MX RT106A MCU, enabling it to run the Alexa voice service directly on the device. It also incorporates the MediaTek MT8516 MPU, which supports voice assistants and machine learning applications.

Raspberry Pi Pico, the high-performance MCU

In this article, we will explore an MCU called Raspberry Pi Pico based on the RP2040 microcontroller chip by Raspberry Pi, and learn how to write a sample code for blinking an LED using CircuitPython. Raspberry Pi Pico is a low-cost, high-performance MCU board that uses the RP2040 chip. This chip boasts a dual-core ARM Cortex-M0+ processor, 264KB of SRAM, and 2MB of flash memory. Additionally, it offers 26 GPIO pins, three analogue inputs, and support for I2C, SPI, and UART protocols. Raspberry Pi Pico can run Python or C/C++ codes, making it remarkably easy to program and customise. Let me guide you through setting up your development environment, writing a simple firmware program, and uploading it to your Raspberry Pi Pico board.